ASSIGNMENT -2

Q.1 Describe different type of cell and outline the structure and function of different component present in them .

Ans. Cells are similar to factories with different labourers and departments that work towards a common objective. Various types of cells perform different functions. Based on cellular structure, there are two types of cells:

- Prokaryotes
- Eukaryotes

Explore: Difference Between Prokaryotic and Eukaryotic Cells

Prokaryotic Cells

Main article: Prokaryotic Cells

1. Prokaryotic cells have no nucleus. Instead, some prokaryotes such as bacteria have a region within the cell where the genetic material is freely suspended. This region is called the nucleoid.
2. They all are single-celled microorganisms. Examples include archaea, bacteria, and cyanobacteria.
3. The cell size ranges from 0.1 to 0.5 µm in diameter.
4. The hereditary material can either be DNA or RNA.
5. Prokaryotes generally reproduce by binary fission, a form of asexual reproduction. They are also known to use conjugation – which is often seen as the prokaryotic equivalent to sexual reproduction (however, it is NOT sexual reproduction).

Eukaryotic Cells

Main article: Eukaryotic Cells

1. Eukaryotic cells are characterised by a true nucleus.
2. The size of the cells ranges between 10–100 µm in diameter.
3. This broad category involves plants, fungi, protozoans, and animals.
4. The plasma membrane is responsible for monitoring the transport of nutrients and electrolytes in and out of the cells. It is also responsible for cell to cell communication.
5. They reproduce sexually as well as asexually.
6. There are some contrasting features between plant and animal cells. For eg., the plant cell contains chloroplast, central vacuoles, and other plastids, whereas the animal cells do not.

Cell Structure

The cell structure comprises individual components with specific functions essential to carry out life's processes. These components include- cell wall, cell membrane, cytoplasm, nucleus, and cell organelles. Read on to explore more insights on cell structure and function.

Cell Membrane

- The cell membrane supports and protects the cell. It controls the movement of substances in and out of the cells. It separates the cell from the external environment. The cell membrane is present in all the cells.
- The cell membrane is the outer covering of a cell within which all other organelles, such as the cytoplasm and nucleus, are enclosed. It is also referred to as the plasma membrane.
- By structure, it is a porous membrane (with pores) which permits the movement of selective substances in and out of the cell. Besides this, the cell membrane also protects the cellular component from damage and leakage.
- It forms the wall-like structure between two cells as well as between the cell and its surroundings.
- Plants are immobile, so their cell structures are well-adapted to protect them from external factors. The cell wall helps to reinforce this function.

Cell Wall

- The cell wall is the most prominent part of the plant's cell structure. It is made up of cellulose, hemicellulose and pectin.
- The cell wall is present exclusively in plant cells. It protects the plasma membrane and other cellular components. The cell wall is also the outermost layer of plant cells.
- It is a rigid and stiff structure surrounding the cell membrane.
- It provides shape and support to the cells and protects them from mechanical shocks and injuries.

Cytoplasm

- The cytoplasm is a thick, clear, jelly-like substance present inside the cell membrane.
- Most of the chemical reactions within a cell take place in this cytoplasm.
- The cell organelles such as endoplasmic reticulum, vacuoles, mitochondria, ribosomes, are suspended in this cytoplasm.

Nucleus

- The nucleus contains the hereditary material of the cell, the DNA.
- It sends signals to the cells to grow, mature, divide and die.
- The nucleus is surrounded by the nuclear envelope that separates the DNA from the rest of the cell.
- The nucleus protects the DNA and is an integral component of a plant's cell structur

Functions of Cell

A cell performs major functions essential for the growth and development of an organism. Important functions of cell are as follows:

Provides Support and Structure

All the organisms are made up of cells. They form the structural basis of all the organisms. The cell wall and the cell membrane are the main components that function to provide support and structure to the organism. For eg., the skin is made up of a large number of cells. Xylem present in the vascular plants is made of cells that provide structural support to the plants.

Facilitate Growth Mitosis

In the process of mitosis, the parent cell divides into the daughter cells. Thus, the cells multiply and facilitate the growth in an organism.

Allows Transport of Substances

Various nutrients are imported by the cells to carry out various chemical processes going on inside the cells. The waste produced by the chemical processes is eliminated from the cells by active and passive transport. Small molecules such as oxygen, carbon dioxide, and ethanol diffuse across the cell membrane along the concentration gradient. This is known as passive transport. The larger molecules diffuse across the cell membrane through active transport where the cells require a lot of energy to transport the substances.

Energy Production

Cells require energy to carry out various chemical processes. This energy is produced by the cells through a process called photosynthesis in plants and respiration in animals.

Aids in Reproduction

A cell aids in reproduction through the processes called mitosis and meiosis. Mitosis is termed as the asexual reproduction where the parent cell divides to form daughter cells. Meiosis causes the daughter cells to be genetically different from the parent cells.

Thus, we can understand why cells are known as the structural and functional unit of life. This is because they are responsible for providing structure to the organisms and perform several functions necessary for carrying out life's processes.

Q.2 Mention the structure and function of chloroplast.

Ans.Chloroplasts are specialized, double-membrane organelles in plants and algae that conduct photosynthesis, converting solar energy into chemical energy (glucose). They contain chlorophyll within a thylakoid system (grana) for light absorption, a dense stroma for carbon fixation, and possess their own DNA and ribosomes.

Microbe Notes +3

Structure of a Chloroplast

- Envelope (Outer & Inner Membranes): The chloroplast is enclosed by a double-membrane envelope. The outer membrane is porous, while the inner membrane is highly selective, regulating molecular traffic.
- Stroma: A thick, alkaline fluid filling the inner space surrounding the thylakoids. It contains enzymes, DNA, ribosomes, and starch granules, and is the site of the Calvin cycle (dark reactions).
- Thylakoid System: A network of interconnected, flattened membrane sacs (thylakoids).
- Grana: Stacks of thylakoids (resembling coins) where light-dependent reactions occur.

- Chlorophyll: A green pigment housed within the thylakoid membranes that absorbs light energy.
- Intermembrane Space: The space between the inner and outer membranes.

Functions of a Chloroplast

- Photosynthesis: The primary function, where light energy is converted into chemical energy and in the thylakoids, which is then used in the stroma to convert into glucose.
- Light Absorption: Chlorophyll captures solar energy.
- Carbon Fixation (Calvin Cycle): The stroma utilizes
- and energy from the light reactions to produce sugars.
- Oxygen Production: Oxygen is released as a byproduct of photolysis (splitting of water).
- Other Metabolic Activities: Chloroplasts are involved in fatty acid synthesis, amino acid synthesis, and immune responses in plants.

Chloroplasts are self-replicating organelles, containing their own genetic material, and are considered to have evolved from ancient cyanobacteria.

Q.3 Mention the structure and function of Protein.

Ans. Proteins are essential, complex macromolecules made of amino acid chains that fold into specific shapes to dictate function, ranging from enzymatic catalysis to structural support. Their structure is organized into four levels—primary (sequence), secondary (folding), tertiary (3D shape), and quaternary (subunit interaction)—which determine their crucial roles in regulation, transport, and immunity.

Structure of Protein

Protein structure is hierarchical, with the sequence of amino acids determining the final 3D conformation.

Primary Structure: The linear, specific sequence of amino acids in a polypeptide chain.

- Secondary Structure: Local folding of the polypeptide chain into regular structures, such as
- -helices and
- -pleated sheets, stabilized by hydrogen bonds.
- Tertiary Structure: The overall three-dimensional shape of a single protein molecule, determined by interactions (hydrophobic, ionic, disulfide bonds) between R groups of amino acids.
- Quaternary Structure: The arrangement and orientation of multiple polypeptide chains (subunits) into a single functional unit.

Functions of Protein

Proteins are vital for almost every cellular process, with functions directly related to their 3D shape.

- Enzymatic Activity: Accelerate metabolic reactions (e.g., enzymes like lipase).
- Structural Support: Provide strength and structure (e.g., collagen in connective tissue, keratin in hair/skin).
- Transport & Storage: Move molecules within the blood (e.g., hemoglobin) or store nutrients (e.g., albumin).
- Immune Protection: Antibodies defend the body against antigens.
- Regulation: Act as hormones to regulate physiological processes (e.g., insulin).
- Movement: Facilitate muscle contraction (e.g., actin and myosin).

Protein denaturation (loss of shape due to heat or pH changes) causes a loss of these functions.